

1 Results

1. Gathered data

(a) Structure of data from Silvics

- i. Angiosperm 79% (98/124) with masting data; Masting and non-masting split almost equally 52% (51/98); Gymnosperm (64 total; 95% (61/64) with masting data; almost all species are masting 89% (54/61)

(b) Number of traits for different hypotheses

- i. For angiosperm, we have data for:
 - A. Predation satiation: seed weight, fruit size, oil content, seed dispersal mode, seed dormancy
 - B. Pollination coupling: pollination mode, reproductive type
 - C. Resource matching: drought tolerance, leaf longevity
- ii. For gymnosperm, we have the same data as for angiosperm, but we only have oil content for masting species and all the gymnosperm species are wind pollinated.

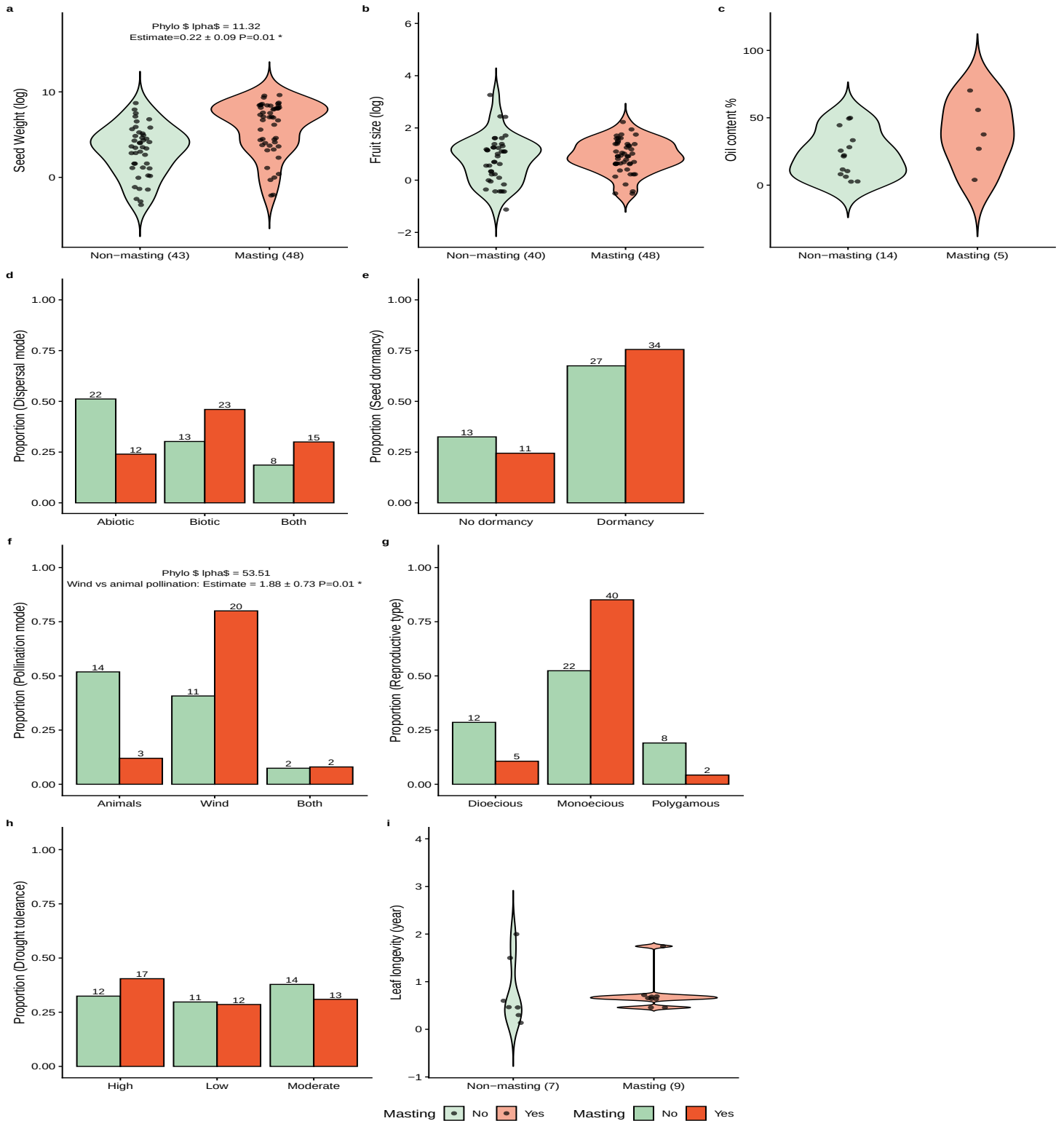


Figure 1: How plant traits related to different hypotheses differ in masting and non-masting species: We show data for traits related to the predator satiation hypothesis (a-e), pollination coupling hypothesis (f-g) and the resource matching hypothesis (h-i) as either violin plots (1-c, i) or proportional histograms (d-h) separated by species categorized as masting (red) or non-masting (green). We show model summaries only where significant (P-value < 0.5) and also report the model sample size (N), regression coefficient (Estimate), and phylogenetic signal parameter α . Full model results are show in supp table XX.

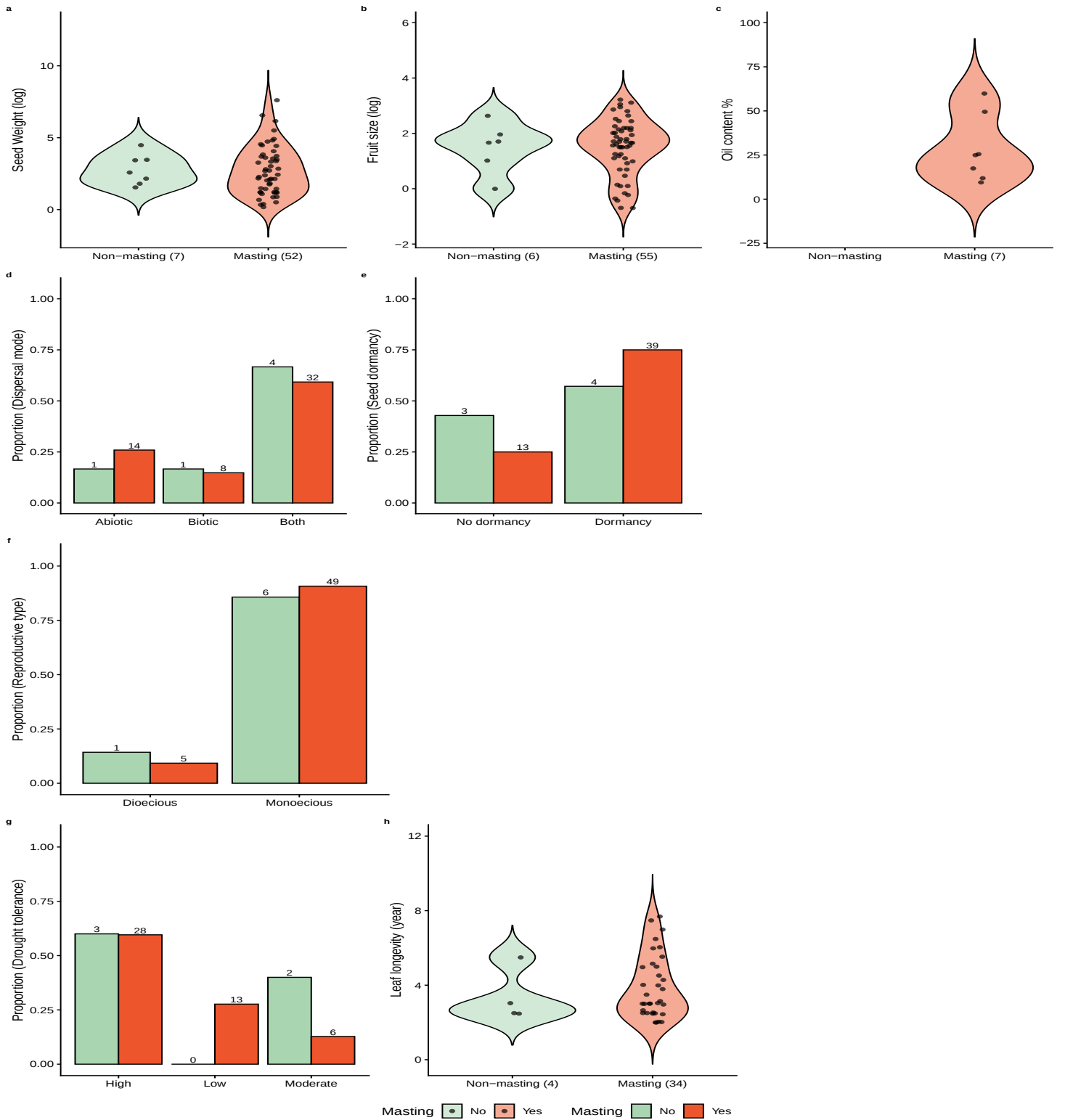


Figure 2: Relationships between plant traits (a) Seed weight; (b) Fruit size; (c) Oil content; (d) Dispersal mode; (e) Seed dormancy; (f) Reproductive mode; (g) Drought tolerance; (h) Leaf longevity and masting based on phylogenetic logistic regression for gymnosperm species. Raw data are shown for each trait: continuous traits are displayed as violin plots, and categorical traits as histograms, the y-axis of the histograms represents the proportion of each category within either the non-mast or mast group. Overlaid model summaries are derived from phylogenetic logistic regression models fitted using phyloglm. For each model, we report sample size (N), regression coefficient (Estimate), phylogenetic signal parameter α , and associated P-value for traits with a P-value < 0.5 . (a)-(e) are traits related to predation satiation hypothesis; (f) is trait related to pollination coupling hypothesis; (g), (h) are traits related to resource matching hypothesis. Legends are shared across panels.

Trait	Predictor	Type	Estimate	SE	P	Phylo α	Half-time	N
Seed weight (log)	logSeedWeight*	Numeric	0.21	0.09	0.02	8.54	0.08	91
Fruit size (log)	logFruit	Numeric	0.08	0.30	0.80	23.25	0.03	88
Oil content (%)	Oil content (%)	Numeric	0.04	0.03	0.19	19.28	0.04	19
Dispersal mode	Biotic vs abiotic	Categorical	0.55	0.53	0.30	8.00	0.09	93
Dispersal mode	Both vs abiotic	Categorical	0.55	0.57	0.33	8.00	0.09	93
Seed dormancy	Dormant vs non-dormant	Categorical	0.03	0.50	0.95	13.07	0.05	85
Pollination mode	Wind vs animal pollination*	Categorical	1.72	0.72	0.02	47.03	0.01	52
Pollination mode	Wind vs animal pollination and animals	Categorical	1.23	1.15	0.29	47.03	0.01	52
Reproductive type	Monoecious vs dioecious	Categorical	0.91	0.68	0.18	15.00	0.05	89
Reproductive type	Monoecious vs polygamous	Categorical	-0.12	0.96	0.90	15.00	0.05	89
Drought tolerance	Low vs high drought tolerance	Categorical	0.14	0.47	0.76	12.57	0.06	79
Drought tolerance	Moderate vs high drought tolerance	Categorical	-0.13	0.46	0.78	12.57	0.06	79
Leaf longevity (years)	Leaf longevity (years)	Numeric	0.26	0.95	0.78	0.90	0.77	16

Table 1: Phylogenetic generalized linear model results for angiosperm species. We modeled each trait separately with masting as a binary variable and traits as either categorical (Type column), where we report (Estimate column) the the difference in log-odds between a given level and its designated baseline reference level, or continuous, where we report the change in log-odds per unit increase of the predictor (Estimate column). We also report the standard error (SE), P-value (with $P < 0.05$ considered significant and marked with an asterisk next to the Predictor), “Phylo α ”, which represents the the phylogenetic signal in the residuals estimated for each trait, and the half-time ($\log(2)/\alpha$), which represents the time required for the influence of ancestral states to decrease by half, where values less than 1 indicating relatively weak phylogenetic signal in the residuals.

Trait	Predictor	Type	Estimate	SE	P	Phylo α	Half-time	N
Seed weight (log)	logSeedWeight	Numeric	-0.03	0.25	0.91	54.43	0.01	59
Fruit size (log)	logFruit	Numeric	0.03	0.36	0.94	54.45	0.01	60
Seed dispersal	Biotic vs abiotic	Categorical	-0.83	0.81	0.30	1.93	0.36	60
Seed dispersal	Both vs abiotic	Categorical	-1.11	0.67	0.10	1.93	0.36	60
Seed dormancy	Dormant vs non-dormant	Categorical	0.71	0.70	0.32	52.15	0.01	59
Reproductive type	Monoecious vs dioecious	Categorical	-1.14	1.02	0.26	0.93	0.74	61
Drought tolerance	Low vs high drought tolerance	Categorical	11.50	94.75	0.90	0.02	36.48	52
Drought tolerance	Moderate vs high drought tolerance	Categorical	-0.25	0.58	0.67	0.02	36.48	52
Leaf longevity (years)	Leaf longevity (years)	Numeric	0.15	0.34	0.67	54.44	0.01	38

Table 2: Phylogenetic generalized linear model results for gymnosperm. Each trait was evaluated in a separate model structured as $\text{Masting} \sim \text{Trait}$, where $\text{masting}(\text{Yes/No})$ is the binary response variable. For categorical traits, the estimate represents the difference in log-odds between a given level and its designated baseline reference level. For numeric traits, the estimate represents the change in log-odds per unit increase of the predictor. SE is the standard error, and P indicates statistical significance, with $P < 0.05$ considered significant and marked with an asterisk (*) in the table. “Phylo α ” represents the phylogenetic signal in the residuals estimated for each trait using phyloglm, we calculated half-time ($\log(2)/\alpha$) to compare with the tree height, which represents the time required for the influence of ancestral states to decrease by half. The tree we used is ultrametric with a standardized height of 1. If half-time is greater than tree height, it indicates strong phylogenetic signal in the residuals; if half-time is smaller than tree height, it indicates relatively weak phylogenetic signal in the residuals. N indicates the number of observations available for each trait.

Group	Trait	Predictor	Estimate	SE	P	Phylo α	Half-time	N
Biotic dispersed	Seed weight (log)	logSeedWeight	0.40	0.15	0.01	54.57	0.01	53
Abiotic dispersed	Seed weight (log)	logSeedWeight	0.12	0.12	0.36	50.33	0.01	36

Table 3: Phylogenetic Generalized Linear Model Results for only seed weight for biotic-dispersed subgroup and abiotic-dispersed subgroup

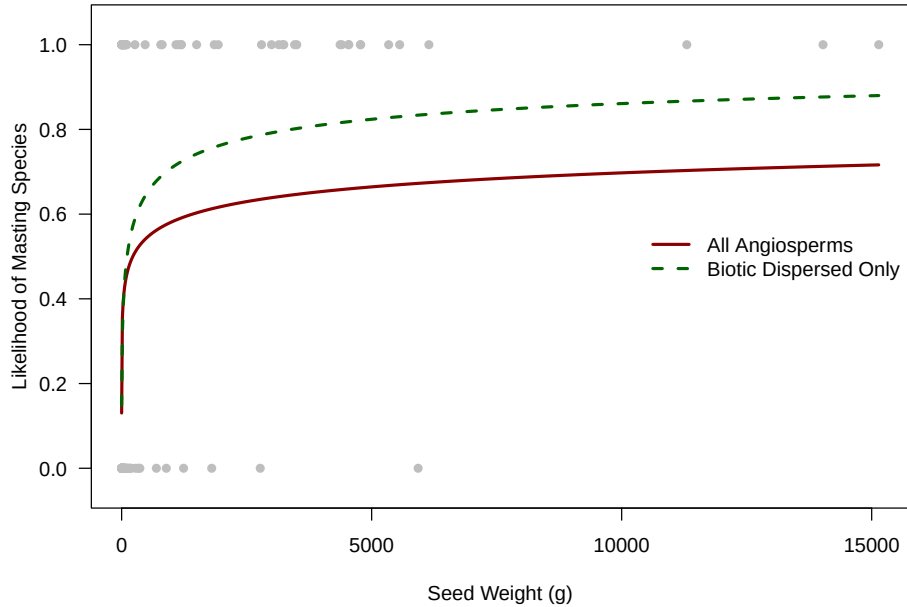


Figure 3: Predicted likelihood of masting as a function of seed weight for all angiosperm species (solid red line) and the biotic-dispersed subgroup (dashed green line), estimated using phylogenetic logistic regression. Individual species observations are plotted as binary data points at 0 (non-masting) and 1 (masting).

1.1 Results for each hypothesis

1. Predation satiation: seed weight only with a small effect and only for angiosperm
 - (a) Only seed weight is related to masting with larger seeds increasing the probability of being a masting species (Figure (1)). For angiosperms only, tripling of seed weight (from a median seed weight of 90.9g to 272.7g) is predicted to increase the likelihood of masting from 0.45 to 0.51 (Figure (3)). And this increase is stronger in biotic-dispersed seed group (Table (3); Figure (3)).
2. Pollination satiation: wind pollination for angiosperm
 - (a) we found wind pollination is related to masting for angiosperm species, wind pollinated species have a significantly higher probability (59%) of being strong masting species. Gymnosperm are all wind pollinated, and gymnosperm has a greater portion of masting species (89% (54/61)).
3. Resource matching: nothing significant

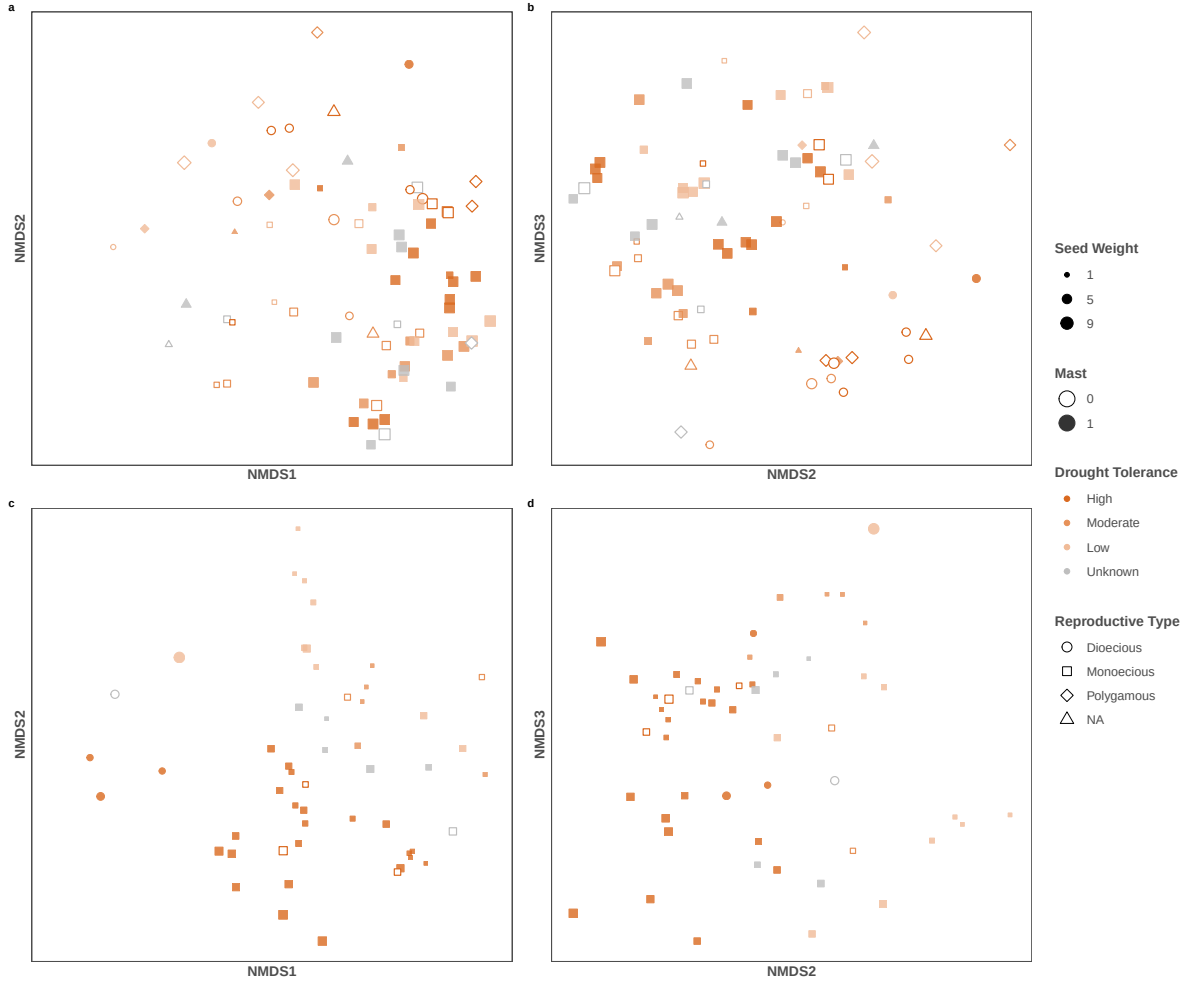


Figure 4: Non-metric multidimensional scaling (NMDS) of trait composition for angiosperm and gymnosperm species. Each point represents a species, with position determined by NMDS axes. Points are colored by drought tolerance, filled by masting or not, shaped by reproductive type, sized by seed weight. Panels are arranged as (a) angiosperm NMDS1–NMDS2, (b) angiosperm NMDS2–NMDS3, (c) gymnosperm NMDS1–NMDS2, and (d) gymnosperm NMDS2–NMDS3.

1.2 Multidimension-NMDS

1. No strong trend when considering single trait, maybe because it happens in multidimensional space
 - (a) We don't see masting aligned well along any axis in the NMDS analysis.
 - (b) We don't have evidence that the combination of traits for any of the hypotheses is driving the pattern of masting.

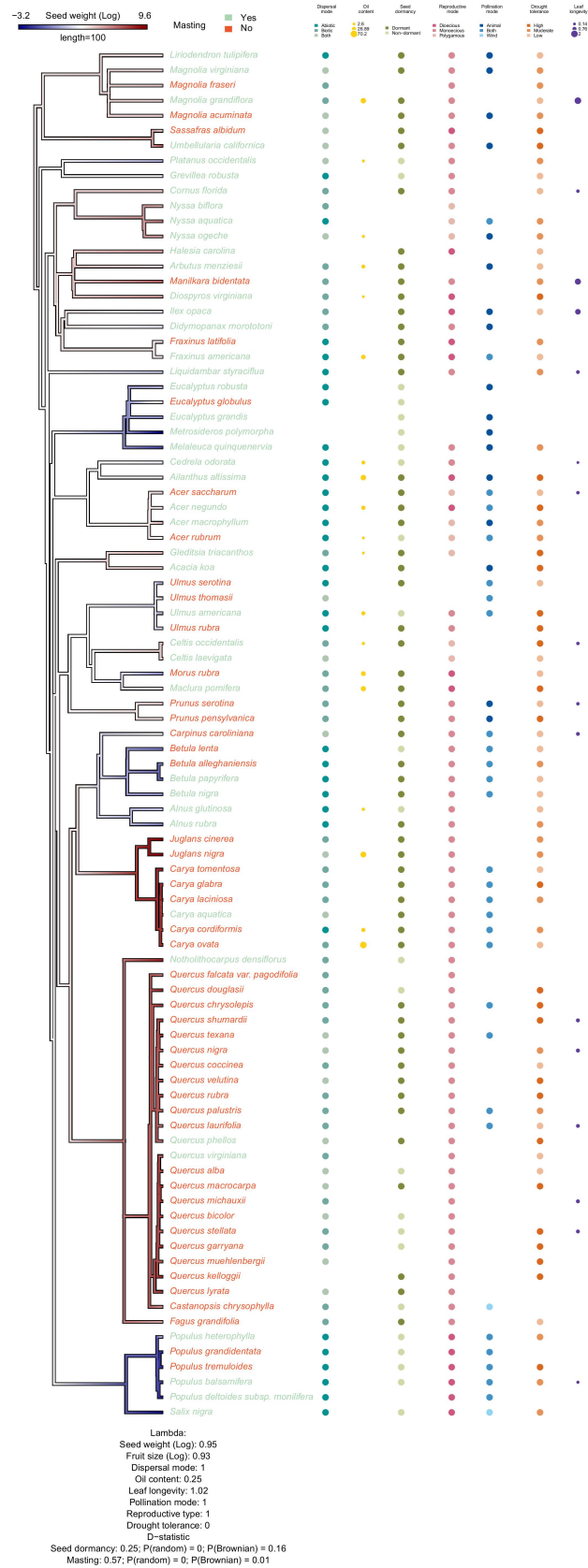


Figure 5: Phylogenetic tree for all angiosperm species included in the analysis. The color scheme of the branch is the trait value for seed weight (log). The color tip is colored by masting or not. The other traits are labeled after the tip labels with different color and size. We also show the predictions of relationship between traits and masting. We calculated the Pagel's λ for continuous traits as well as multilevel categorical traits, and calculated D-statistic for binary traits

Group	Trait	(λ)
Angiosperm	Seed weight (Log)	0.95
Angiosperm	Fruit size (Log)	0.93
Angiosperm	Oil content	0.25
Angiosperm	Leaf longevity	1.00
Angiosperm	Dispersal mode	1.00
Angiosperm	Pollination mode	1.00
Angiosperm	Reproductive type	1.00
Angiosperm	Drought tolerance	0.00
Gymnosperm	Seed weight (Log)	0.97
Gymnosperm	Fruit size (Log)	0.90
Gymnosperm	Oil content	0.00
Gymnosperm	Leaf longevity	0.90
Gymnosperm	Dispersal mode	1.00
Gymnosperm	Drought tolerance	1.00

Table 4: Phylogenetic Signal (Pagel’s λ) for continous traits and multi-level categorical traits. $\lambda = 0$ indicates no phylogenetic signal, trait districuted independent of the phylogeny; $0 < \lambda < 1$ means moderate phylogenetic signal, trait is partially structured by the phylogeny; $\lambda = 1$ indicates strong phylogenetic signal, trait follows Brownian motion.

Group	Trait	Estimated D	P(random)	P(Brownian)
Angiosperm	Seed dormancy	0.25	0	0.16
Angiosperm	Masting	0.58	0	0.01
Gymnosperm	Masting	0.51	0.02	0.10
Gymnosperm	Seed dormancy	0.98	0.41	0
Gymnosperm	Reproductive type	-0.52	0	0.89

Table 5: Phylogenetic Signal (D-statistic) for the phylogenetic signal of binary traits, with probabilities calculated under a random phylogenetic structure and under a Brownian motion phylogenetic structure. $D = 1$ indicates the trait is randomly distributed across the phylogeny; $D = 0$ indicates the trait follows a Brownian motion model; $0 < D < 1$ indivates moderate phylogenetic signal; $D < 0$ indicates the trait is more clumped than expected under Brownian motion; and $D > 1$ indicates overdispersion. P random is the p-value for the null hypothesis that the trait is randomly distributed across the phylogeny (no phylogenetic signal), and P Brownian is the p-value for the null hypothesis that the trait follows a Brownian motion model (strong phylogenetic signal). A low p-value indicates rejection of the corresponding null hypothesis.

1.3 Phylogenetic patterns

1. Pagel's λ for continuous traits and multilevel categorical traits in Table 4, D-statistic for binary traits in Table 5.
 - (a) Mastig shows a moderate phylogenetic signal in both gymnosperms (estimated $D = 0.51$) and angiosperms (estimated $D = 0.58$) (Table 5).
 - (b) Seed weight has strong phylogenetic signal for both angiosperm ($\lambda = 0.95$) and gymnosperm ($\lambda = 0.97$). We can see the similarity in certain group in Figure 5 it's relatively reliable estimate. Dispersal mode, leaf longevity, pollination mode and reproductive type also showed high phylogenetic signal in Table 4, but be cautious that λ for categorical traits might not be accurate.
 - (c) Be cautious that λ might not be comparable among traits, not only because the way λ is calculated for continuous traits and categorical traits are different, but also because λ might depend on the data structure when sampling is uneven, take leaf longevity as an example in Figure 5, leaf longevity data is sparse and nested in *Quercus*. (5/16).